Medicine Recommendation System

Project Report

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COURSE: Cyber Security and Digital Forensics

YEAR: 1st Year — VIT Bhopal

DATE: 30 September 2026

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1. Problem Statement

In daily life, people often experience symptoms but are unsure about what disease they might indicate. Many individuals rely on random internet searches, which can lead to confusion, misinformation, or panic. There is a lack of simple tools that can help users understand possible health conditions based on symptoms in a structured and reliable way.

This project aims to build a system that takes user symptoms as input and predicts the most likely disease using machine learning, along with recommending basic medicines and precautions.

2. Why This Problem Matters

This problem is relevant because healthcare awareness is still limited, especially among students and young individuals who may ignore early symptoms. A simple AI-based system can act as a first-level guide and help users make more informed decisions.

The goal is not to replace doctors, but to assist users by providing initial insights and encouraging them to seek proper medical advice when necessary.

3. Objective

- To build a system that predicts diseases from user symptoms
- To suggest medicines and basic precautions
- To apply machine learning for real-world problem solving
- To create a simple and interactive user interface
- To understand the complete ML workflow from data to prediction

4. Proposed Solution

- Develop a machine learning-based system that takes user symptoms as input
- Convert symptoms into a structured format for model processing
- Use a Decision Tree and Random Forest model to predict the most likely disease
- Retrieve relevant medicines and precautions from a dataset
- Display results in a clear and user-friendly format
- Provide an interactive system that allows users to recheck symptoms easily

5. Approach and Methodology

Data Collection:

Two datasets were used: Disease-symptom dataset (from Kaggle) and a Medicine and precautions dataset.

Data Preprocessing:

- Cleaned inconsistent column names
- Handled missing and incorrect values
- Converted symptom data into binary format (0/1)
- Ensured uniform formatting for disease names

Feature Engineering:

Symptoms were converted into binary feature vectors where 1 indicates the symptom is present and 0 indicates it is absent. This allowed the machine learning model to process the data effectively.

Model Selection & Comparison

A Decision Tree Classifier and Random Forest Classifier were compared because they work exceptionally well with categorical/binary features and provide good interpretability.

Model	Accuracy
Decision Tree	~95%
Random Forest	~97.56%

Random Forest performed better, so that is the final model saved and deployed in the application.

6. Key Features

- Disease prediction based on binary symptom vectors
- Smart medicine and precaution recommendation system
- Confidence score comparison chart for top matches
- Interactive Command Line Interface (CLI)
- Option to export the full diagnosis report as a text file

7. Key Decisions Made

1. **Choosing a Real-Life Problem:** Focused on healthcare access in small towns for maximum practical impact.
2. **Using Random Forest:** Selected over a simple Decision Tree due to higher precision and better generalization.
3. **Keeping Interface Simple:** Used a clean CLI environment for faster terminal execution and easy deployment.

8. Challenges Faced

- Resolving dataset inconsistencies and structural formatting issues inside the CSV files.
- Vectorizing raw text symptoms into multi-dimensional numerical inputs dynamically.
- Accurately matching predicted diseases with the corresponding medicines dataset without cross-mapping errors.

9. Results and Observations

- The system successfully predicts diseases based on user-input symptom numbers.
- The confidence levels effectively help compare multiple possible diseases if symptoms overlap.
- The entire solution demonstrates a practical, working application of machine learning pipelines.

10. Limitations

- Limited dataset size spanning a fixed number of health conditions.
- Lacks a native web or graphical interface (GUI).
- **Crucial:** It is purely informational and not a replacement for professional medical advice.

11. What I Learned

Technical Skills: Python programming, pipeline preprocessing using scikit-learn, dataset engineering with pandas, and model serialization using pickle.

Practical Insights: Understood how data quality directly dictates model performance and how binary feature vectors enable text classification.

12. Conclusion

The Disease Prediction and Medicine Recommendation System successfully addresses the problem of identifying potential conditions from basic symptoms. It bridges the gap between raw machine learning theory and an interactive utility tool.

13. References

- Python Documentation & Standard Libraries
- Scikit-learn Documentation (Random Forest / Decision Trees)
- Pandas & NumPy Data Analysis Reference Guides